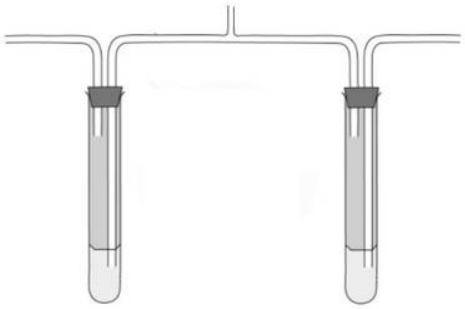
Type MS · Explain how respiration releases energy and ATP transfers it

Answer 1 of 7 · Question 7 part b · 5 marks

2024 JUN · Paper 02 (regional R) · find part b in this mark scheme extract

7 (a)	(i)	glucose ⟶ lactic acid/lactate;	ignore any ref to CO_2 and H_2O on right side of equation	(1)
	(ii)	$C_6H_{12}O_6 + 6O_2; \longrightarrow 6H_2O + 6CO_2;$		(2)
	(iii)	general diagram; tubing correct way round; 	labels not needed long short, long short short long, short long	(2)
	(iv)	• inhale and exhale/breathe through the mouthpiece; • at rest; • time how long it takes the limewater to go cloudy/milky; • repeat following exercise;		(4)
(b)		any five from • energy released from (breakdown of) glucose; • during (aerobic) respiration; • (energy used) to add phosphate group; • to convert ADP to form ATP; • energy is released from ATP breakdown of ATP releases energy; • breaking of phosphate bond/ATP hydrolysed/ATP breaks down into ADP + P(i); • to provide energy for muscle contraction;		(5)

Total for Question 7 = 14 marks

Type MS · Explain how respiration releases energy and ATP transfers it

Answer 2 of 7 · Question 3 part b.iii · 2 marks

2020 JAN · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

3 (a) (i)	**Structure** — **Function** cell membrane → helps the cell to move ribosomes → make proteins flagellum → holds the genetic code controls which substances enter and exit the cell → protection against pathogens	Reject more than one line from each structure One mark for each correct line	3
(ii)	Any two from: • (bacterial cell) has a cell wall (1) • (bacterial cell) does not have a nucleus (1) • (bacterial cell) has a plasmid (1)		2
(b) (i)	**The only correct answer is B diffusion** *A is incorrect because it is down a diffusion gradient* *C is incorrect because it is not removal from the bacterium* *D is incorrect because osmosis only involves movement of water*		1
(ii)	• glucose (1) • carbon dioxide (1)	Answers must be in the correct order Allow correct chemical formulae	2
(iii)	Any two from: • to provide energy (1) • for movement (1) • for growth/reproduction/divide/mitosis (1) • for chemical reactions/producing organelles/named organelle (1)		2

Type MS · Explain how respiration releases energy and ATP transfers it

Answer 2 of 7 · Question 3 part b.iii · 2 marks

2020 JAN · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

Total for Question 3 = 10 marks

Type MS · Explain how respiration releases energy and ATP transfers it

Answer 3 of 7 · Question 7 part a.i · 1 mark

2020 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

7 (a) (i)		aerobic respiration (1)		1
	(ii)	• shorter (1) • fatter/wider/thicker (1)		2
	(iii)	• shortening causes tension/pull (1) • tendon inelastic (1) • pulls on radius/bone (1)		3
(b)		• ball and socket indicated labelled bone (1) • capsule/ligament labelled (1) • cartilage labelled (1) • synovial fluid labelled (1) • synovial membrane labelled (1) • overall appearance of diagram (1)		6

Total 12 marks

Type MS · Explain how respiration releases energy and ATP transfers it

Answer 3 of 7 · Question 7 part a.i · 1 mark

2020 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how respiration releases energy and ATP transfers it

Answer 4 of 7 · Question 5 part a.ii · 2 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number					
5	(a)	(i)	$C_6H_{12}O_6$; $6H_2O$;		2
		(ii)	ATP breaks down/hydrolysed; to ADP and Pi;	Allow phosphate/P	2
	(b)	(i)	• suitable axis labels with units; • time on x axis; • suitable scale; • correct plots;		4
		(ii)	• suitable curve;		1
		(iii)	Five from; • mass of haemoglobin increases (with time/training at altitude); • increased number of red blood cells/more haemoglobin in the blood; • haemoglobin binds oxygen/red blood cells carry oxygen; • more oxygen transported around the body/in the blood; • to muscle (cells/tissue); • for aerobic respiration; • to release energy;		5

Total 14 marks

Type MS · Explain how respiration releases energy and ATP transfers it

Answer 4 of 7 · Question 5 part a.ii · 2 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how respiration releases energy and ATP transfers it

Answer 5 of 7 · Question 4 part a.iv · 1 mark

2023 JUN · Paper 02 · extract 1 of 2 · find part a.iv in this mark scheme extract

Number					
4	(a)	(i)	• glucose; • carbon dioxide and water;	Reject formulae	2
		(ii)	aerobic;		1
		(iii)	B; (mitochondrion)		1
		(iv)	ATP/adenosine triphosphate		1
	(b)	(i)	Any two from • more people/repeat tests: • different gender/diet/location/age; • carry out test over more years/longer time period; • calculate the mean/ average;		2
		(ii)	• 7250 - 6550;	ecf if wrong figures used in a subtraction calculation but correct answer 1 mark	
			• = 700kJ;		2
		(iii)	Any five from • metabolic rate increases in colder temperatures; • correct comparative reading from graph e.g. metabolic rate is higher in January compared to July; • (in colder temperatures) increased rate of respiration; • releases heat (energy)/more energy needed; • (energy needed) to maintain body temperature/reference to keeping body warm; • (in warmer weather) more heat lost (by radiation / convection/sweating);	ORA for mps 1, 3, 4, 5, 6 Allow marking point if correct figures given from graph	5
		(iv)	uptake of oxygen/carbon dioxide output;	Allow amount of glucose used	1

Total 15

Type MS · Explain how respiration releases energy and ATP transfers it

Answer 5 of 7 · Question 4 part a.iv · 1 mark

2023 JUN · Paper 02 · extract 2 of 2 · find part a.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how respiration releases energy and ATP transfers it

Answer 6 of 7 · Question 4 part b.iii · 5 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part b.iii in this mark scheme extract

Number					
4	(a)	(i)	• glucose; • carbon dioxide and water;	Reject formulae	2
		(ii)	aerobic;		1
		(iii)	B; (mitochondrion)		1
		(iv)	ATP/adenosine triphosphate		1
	(b)	(i)	Any two from • more people/repeat tests: • different gender/diet/location/age; • carry out test over more years/longer time period; • calculate the mean/ average;		2
		(ii)	• 7250 - 6550;	ecf if wrong figures used in a subtraction calculation but correct answer 1 mark	
			• = 700kJ;		2
		(iii)	Any five from • metabolic rate increases in colder temperatures; • correct comparative reading from graph e.g. metabolic rate is higher in January compared to July; • (in colder temperatures) increased rate of respiration; • releases heat (energy)/more energy needed; • (energy needed) to maintain body temperature/reference to keeping body warm; • (in warmer weather) more heat lost (by radiation / convection/sweating);	ORA for mps 1, 3, 4, 5, 6 Allow marking point if correct figures given from graph	5
		(iv)	uptake of oxygen/carbon dioxide output;	Allow amount of glucose used	1

Total 15

Type MS · Explain how respiration releases energy and ATP transfers it

Answer 6 of 7 · Question 4 part b.iii · 5 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how respiration releases energy and ATP transfers it

Answer 7 of 7 · Question 4 part b.iv · 1 mark

2023 JUN · Paper 02 · extract 1 of 2 · find part b.iv in this mark scheme extract

Number					
4	(a)	(i)	• glucose; • carbon dioxide and water;	Reject formulae	2
		(ii)	aerobic;		1
		(iii)	B; (mitochondrion)		1
		(iv)	ATP/adenosine triphosphate		1
	(b)	(i)	Any two from • more people/repeat tests: • different gender/diet/location/age; • carry out test over more years/longer time period; • calculate the mean/ average;		2
		(ii)	• 7250 - 6550;	ecf if wrong figures used in a subtraction calculation but correct answer 1 mark	
			• = 700kJ;		2
		(iii)	Any five from • metabolic rate increases in colder temperatures; • correct comparative reading from graph e.g. metabolic rate is higher in January compared to July; • (in colder temperatures) increased rate of respiration; • releases heat (energy)/more energy needed; • (energy needed) to maintain body temperature/reference to keeping body warm; • (in warmer weather) more heat lost (by radiation / convection/sweating);	ORA for mps 1, 3, 4, 5, 6 Allow marking point if correct figures given from graph	5
		(iv)	uptake of oxygen/carbon dioxide output;	Allow amount of glucose used	1

Total 15

Type MS · Explain how respiration releases energy and ATP transfers it

Answer 7 of 7 · Question 4 part b.iv · 1 mark

2023 JUN · Paper 02 · extract 2 of 2 · find part b.iv in this mark scheme extract

Question number	Answer	Notes	Marks